INSURESCOPE

COMPREHENSIVE ANALYSIS OF MEDICAL INSURANCE COSTS AND KEY COST DRIVING FACTORS

Project Report – Stage 4: Documentation, Insights and Presentation

Tools used : Python, Pandas, Matplotib ,Seaborn and Power BI

Data Set: Medical cost Personal Data set

Dataset Size:1,338 records and 7 columns

Project objective

- The objective of InsureScope is to analyse medical insurance costs and identify the key factors associated with variations in insurance charges. The project uses Python for exploratory data analysis and Power BI for interactive data visualization.
- The analysis focuses on age, BMI, smoking status, gender, number of children, and region to understand their relationship with insurance costs.

DATASET DESCRIPTION

The Medical Cost Personal Dataset contains 1,338 records and seven columns.

-Age – Age of the individual.

- Sex – Gender of the individual.
- BMI – Body Mass Index.
- Children – Number of dependants.
- Smoker – Smoking status.
- Region – Residential region.
- Charges – Medical insurance cost.

The dataset was used to explore customer characteristics and their association with medical insurance charges.

POWER BI DASHBOARD

The Power BI dashboard aligns all charts on a single report page by connecting Python to Power BI. First, load and prepare the medical insurance dataset in Power Query, then enable Python under Power BI Desktop settings by selecting the installed Python distribution. Python scripts using Pandas, Matplotlib and Seaborn can be added through the Python script connector or used in Python visualizations to create charts that appear alongside native Power BI visuals. Use a consistent dashboard grid with matching visual dimensions, aligned edges, clear titles, coordinated colours and sufficient spacing so that every chart is easy to compare. Recommended visuals include a KPI card for average charges, a smoker-versus-non-smoker comparison, an age-and-charges scatter

plot, BMI distribution charts, and demographic comparisons by sex, region and number of children. Add slicers for smoker status, sex, region and age group, and configure visual interactions so that all charts respond consistently to user selections. Before publishing, verify that Python-generated visuals and native Power BI charts use the same data fields, filters and aggregation rules



KEY INSIGHTS:

1. Smoking Status: Smokers tend to have substantially higher insurance charges than non-smokers.
2. Age: Insurance charges generally show an increasing trend with age.

3. BMI: BMI may be associated with variations in medical insurance charges.
4. Sex: Comparing charges by sex helps identify possible differences between groups.
5. Region and Children: These variables provide additional information about differences in insurance costs.

INTERPRETATION AND FINDINGS:

The analysis suggests that smoking status is an important factor associated with medical insurance charges. Age and BMI may also contribute to differences in costs. Regional differences, sex and number of children provide additional perspectives. These relationships indicate associations, not necessarily direct causes

BUSINESS RECOMMENDATIONS:

- Consider smoking status when studying insurance cost variations.
- Analyse age and BMI together with other customer characteristics.
- Compare average charges and distributions across different groups.
- Investigate unusually high charges to understand their possible reasons.
- Use additional data and appropriate validation before making insurance pricing decisions.

FINAL STORY OF THE DASHBOARD:

InsureScope transforms a medical insurance dataset into meaningful insights using Python-based EDA and Power BI visualizations. The analysis highlights differences in charges across customer groups, with smoking status emerging as an important factor. Age and BMI provide further context, while regional and demographic comparisons complete the analysis.

The dashboard makes these findings easier to understand and communicate, supporting further investigation and data-informed decisions.

TYPES OF ANALYSIS:

1. Descriptive Analysis

Descriptive analysis summarizes the medical insurance dataset using statistical measures such as mean, median, count and standard deviation. In this project, it helps understand the average insurance charges, age distribution, BMI and smoking status of customers.

2. Diagnostic Analysis

Diagnostic analysis examines relationships between variables to understand why insurance charges differ. In this project, charges are compared across smokers and non-smokers, age groups, BMI levels and regions to identify possible reasons for cost variations.

3. Predictive Analysis

Predictive analysis uses statistical techniques and machine learning algorithms to estimate future or unknown outcomes. In this project, a regression model could be developed to predict medical insurance charges based on age, BMI, smoking status and other variables. This requires a separate predictive modelling step.

4. Prescriptive Analysis

Prescriptive analysis suggests suitable actions based on analytical findings. In this project, recommendations include investigating high-cost cases, analysing smoking-related cost differences and using additional data to support insurance cost management decisions.

FUTURE ENHANCEMENT:

- Develop a machine-learning model to predict medical insurance charges.
- Add interactive slicers and KPI cards to the Power BI dashboard.
- Perform statistical tests to validate relationships between variables.
- Include more relevant healthcare and customer information.
- Improve the dashboard with additional comparisons and visualizations.

CONCLUSION:

- These four types of analysis help transform raw medical insurance data into meaningful insights. The current project primarily focuses on descriptive and diagnostic analysis, while predictive and prescriptive analysis are potential future enhancements.
- The InsureScope project demonstrates how Exploratory Data Analysis (EDA) can transform raw medical insurance data into meaningful insights. Python was used for data cleaning, analysis and visualization, while Power BI was used to present the findings in a dashboard.
- The project identifies smoking status, age and BMI as important factors associated with insurance cost variations. Overall, this project improves understanding of medical insurance charges and demonstrates the value of da analytics in business decision-making.